1.9 Logarithmic functions

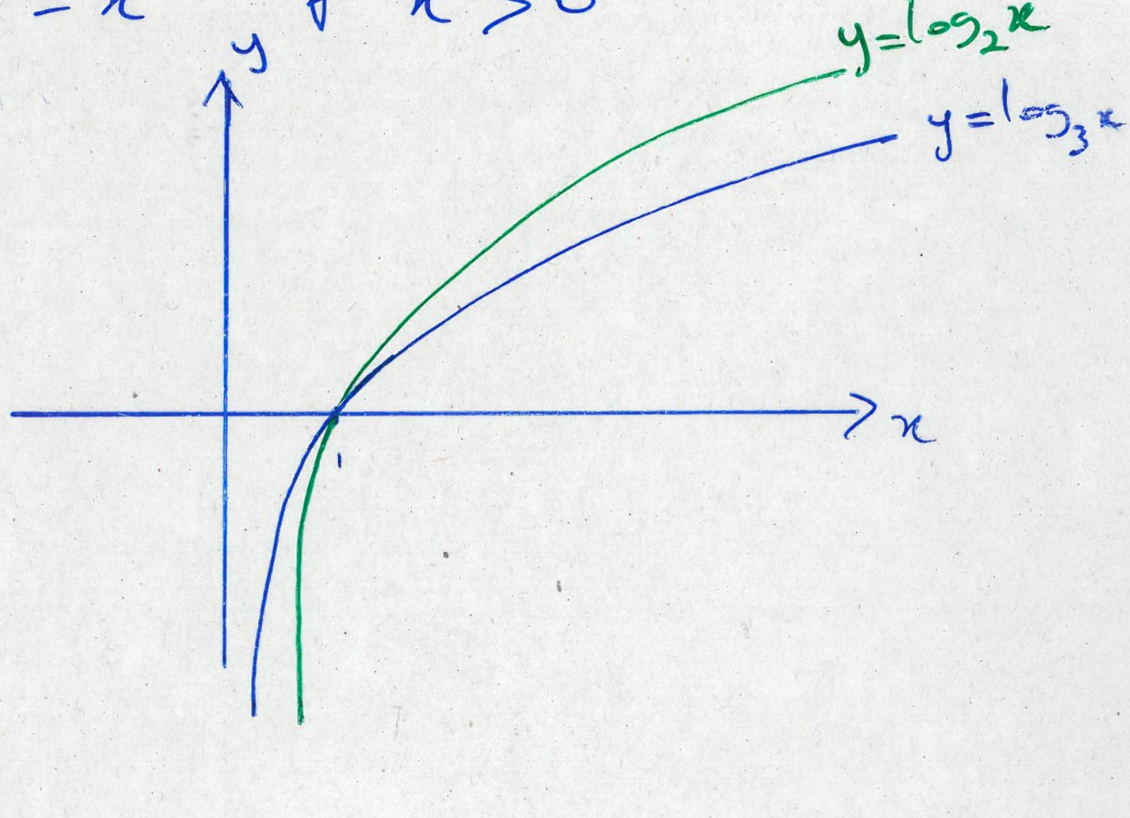
If $a > 0$ and $a \neq 1$, the inverse of $f(x) = a^x$ is $f^{-1}(x) = \log_a(x)$ and is called the logarithmic function with base a.

Therefore,

$$\log_a x = y \iff a^y = x$$

$$\log_a(a^x) = x \quad \forall x \in \mathbb{R}$$

$$a^{\log_a x} = x \quad \forall x > 0$$



Laws of logarithms :

$$\log_a(xy) = \log_a x + \log_a y$$

$$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$$

$$\log_a(x^r) = r.\log_a x \quad , \quad r \in \mathbb{R}$$

Natural Logarithms : The Logarithm with base "e" is called the natural logarithm

$$\log_e x = \ln x$$

$$\ln x = y \iff e^y = x$$

$$\ln(e^x) = x \quad , \quad \forall x \in \mathbb{R}$$

$$e^{\ln x} = x \quad , \quad \forall x > 0$$

In particular $\quad \ln e = 1$

Example.1 : Solve for x

$$e^{5-3x} = 10$$

Change of base formula : For any positive number a $(a \neq 1)$, we have

$$\log_a x = \frac{\ln x}{\ln a}$$

Example 1

$$e^{5-3x} = 10 \Rightarrow \ln\left(e^{5-3x}\right) = \ln 10$$

$$\Rightarrow \quad 5-3x = \ln 10$$

$$\Rightarrow \quad x = \frac{1}{3}(5 - \ln 10)$$

Example 2: Express $\log_3 5$ using the natural logarithm

$$\therefore \quad \log_3 5 = \frac{\ln 5}{\ln 3}$$